Qwen2.5-VL-3B on Mac mini (MLX): test findings

Date: 2026-10-23 · IDP web extraction (PAN, Aadhaar) · Status: findings only, no decision yet

Setup

Machine	Mac mini "mini3" (10.0.3.66), 24 GB unified memory
Model	Qwen2.5-VL-3B-Instruct, 4-bit (MLX)
Engine	MLX `mlx_vlm.server` 0.7.10, continuous batching (`--max-num-seqs 8`)
Test	Same as the Qwen3.5-4B tests: 10 real documents (4 PAN, 6 Aadhaar), 20 requests per level, 1 / 2 / 4 / 8 users at the same time, every request a unique copy of its image, same prompts
Compared with	Qwen3.5-4B on MLX, same machine and test (see LLAMACPP_VS_MLX.md)

"Typical" = median wait (half the users waited less). "Slow" = 95th percentile (1 in 20 waited longer).

Two runs

Qwen2.5-VL cuts an image into 28 px pieces, Qwen3.5 into 32 px pieces, so the same image gives Qwen2.5-VL about 25-35% more tokens to read.

- **Run 1, same images:** the 3B saw exactly the same images as the 4B (more tokens for the 3B).
- **Run 2, same tokens:** images scaled by 0.875 (= 28/32) for the 3B, so it read exactly as many image tokens as the 4B (checked per image: e.g. 669 vs 668, 1113 vs 1113).

Wait per user

Users at the same time	3B run 1 typical / slow	3B run 2 typical / slow	4B MLX typical / slow
1	3.6 s / 7.0 s	3.2 s / 6.3 s	3.9 s / 7.4 s
2	6.2 s / 11.3 s	5.6 s / 9.9 s	6.1 s / 11.0 s
4	11.7 s / 17.2 s	9.6 s / 19.5 s	9.9 s / 19.7 s
8	23.4 s / 45.7 s	18.9 s / 36.2 s	21.9 s / 40.6 s

Throughput, accuracy and errors

Users at the same time	3B run 1 docs/min	3B run 2 docs/min	4B MLX docs/min	3B run 1 accuracy	3B run 2 accuracy	4B MLX accuracy
1	12.5	15.7	13.0	84.0%	87.5%	100%
2	15.1	19.1	16.7	84.9%	84.8%	99.1%

Users at the same time	3B run 1 docs/min	3B run 2 docs/min	4B MLX docs/min	3B run 1 accuracy	3B run 2 accuracy	4B MLX accuracy
4	15.8	20.2	18.8	83.2%	84.8%	99.1%
8	16.9	18.8	19.1	84.8%	85.0%	99.1%

Failed requests (answer was not valid JSON): run 1: 3 of 80 (at 1, 2 and 4 users) · run 2: 1 of 80 (at 8 users) · 4B: none.

Accuracy = fields matching the correct values (56 fields per round of the 10 images).

Where the 3B missed

Field	Run 1 misses	Run 2 misses
Aadhaar number	28	18
Address	25	28
Name (Aadhaar)	0	10
Date of birth	8	0
PIN code	7	8
PAN fields	0	0

All misses were on Aadhaar documents (cards and letters); PAN was read correctly every time on both runs.

Observations

- Matching the token count made the 3B faster than the 4B at every level: 3.2 vs 3.9 s for one user, 18.9 vs 21.9 s at 8 users, up to 20 documents per minute.
- Its accuracy was about 85% in both runs, against 99-100% for the 4B.
- The accuracy was the same with 1 user as with 8 (84-88%), so the misses are not caused by load: they appear even when the model handles one request at a time.
- PAN results matched the 4B; the gap is entirely on Aadhaar (number, address, name, PIN code).
- MLX does not enforce the output format for this setup (see LLAMACPP_VS_MLX.md), which may explain the few invalid answers.

Limits

- Only PAN and Aadhaar were tested (10 images). Driving licence is not tested yet.
- One machine: Mac mini .66. Other machines will give different numbers.
- Prompts were written and tuned on the Qwen3.5-4B; a prompt tuned for the 3B might change its accuracy.